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# An Artificial Intelligence Program to Advise Physicians Regarding Antimicrobial Therapy*

EDWARD H. SHORTLIFFE†, STANTON G. AXLINE, BRUCE G. BUCHANAN, THOMAS C. MERIGAN, AND STANLEY N. COHEN

*Stanford University, Stanford, California 94305*

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An antimicrobial therapy consultation system has been developed which utilizes a flexible representation of knowledge. The novel design facilitates interactive advice-giving sessions with physicians. An ability to display reasons for making decisions at the request of the user permits the program to serve a tutorial as well as consultative role. The feasibility of the judgmental rule approach which the program uses has been demonstrated with a limited knowledge base of approximately 100 rules. Its ultimate success as a clinically useful tool depends upon acquisition of additional rules and thus upon co-operation of infectious disease experts willing to improve the program's knowledge base. The techniques for acquisition, representation, and utilization of knowledge, plus considerations of natural language processing, draw upon and contribute to current Artificial Intelligence research.

## INTRODUCTION

Artificial Intelligence (AI) research can contribute to medical practice and instruction because of its emphasis on using judgmental knowledge for the solution of complex problems. The foundations for AI research were well defined by the early 1960's (*16*), but until recently little effort was directed specifically toward applying AI techniques to medical problems. Within the last several years, however, programs have been developed which undertake such diverse tasks as simulating a paranoid patient during a psychiatric interview (*4*), answering questions about a complex neurological data base (*9*), discussing clinical findings and giving advice regarding the diagnosis of glaucoma (*11*, *12*), and aiding in the formation of theory to explain experimental neurophysiological findings (*17*).

This communication describes a therapeutic consultation system which has been developed at the Stanford University School of Medicine. The principal goal of the

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† To whom requests for reprints should be sent. Address: Division of Clinical Pharmacology, Department of Medicine, Stanford University School of Medicine, Stanford, California 94305.

project has been to devise a computer-based system for assisting and educating physicians who need advice about appropriate antimicrobial therapy. In addition, the program is of interest to computer scientists because of its novel organization of clinical, pharmacological, and microbiological knowledge, its methods for acquisition of new inference rules, its use of the knowledge to give advice, and its ability to explain the reasons for its decisions.

## Background

The basis of rational antimicrobial therapy decisions is identification of the micro-organisms causing the infectious disease. Accurate identification is important because drugs which are highly effective against certain organisms are often useless against others. The patient's clinical status and history, including such information as previous infections and treatments, provide valuable identifying data for the physician. However, bacteriological cultures that use specimens taken from the site of the patient's infection usually provide the most definitive identifying information.

Initial culture reports from a microbiology laboratory may become available within 12 hours from the time a clinical specimen is obtained from the patient. The information in these early reports often serves to classify the organism in general terms but does not permit precise identification. It may be clinically unwise to postpone therapy until identification of the infecting organism can be made with certainty, a process that usually requires 24–48 hours or longer. Thus, it is necessary for the physician to estimate the range of possible organisms and to start appropriate treatment even before the laboratory is able to identify the offending organism and its antibiotic sensitivity.

Studies have shown that physicians will often reach therapeutic decisions which differ significantly from those which would have been suggested by infectious disease experts practicing at the same institution (*5*, *14*). Nonexperts sometimes choose a drug regimen designed to cover for all possibilities, prescribing either several drugs or one of the so-called "broad spectrum" antibiotics even though appropriate utilization of clinical clues might have led to a more rational (and often less toxic) therapy. Amid a hospital environment in which professional resources are often overburdened, a computer-based system that could serve effectively in a consultation role to the nonexpert—and gain his respect—would be highly useful. The computer system described here has been designed to provide a readily accessible consultative and instructive program which will help bridge this gap between practicing physicians and experts in infectious disease therapy.

## Strategy

The strategy of this project has been to design a computer-based system that contains much of the knowledge of an expert in infectious disease therapy. Such a program should be able to give interactive consultations to nonexperts and explain reasons for intermediate and final decisions when queried. An alternative approach would be to build a statistical model of decision making in this clinical field. We felt it desirable to follow the consequences of chained judgmental rules, however, and therefore we did not adopt a purely statistical approach.

Our program is given its knowledge of infectious disease therapy in much the same way that a medical student learns about the subject. During the initial period of its development, the consultation system has been given a "classroom knowledge" of antimicrobial therapy which might be compared to the basic science knowledge of a medical student prior to his first clinical clerkships in the patient-care environment. The program knows certain fundamental information about most antimicrobial drugs and clinically important bacteria, and we have given it approximately 100 decision rules which it can use to provide rather rudimentary advice about the treatment of infectious diseases.

The next stage in the development of the program, a period analogous to the clinical years of the medical student, will involve challenging the system with therapeutic problems. Experts on infectious disease therapy will monitor the program's performance under such circumstances, and will then modify its rules in order to help it become a more adequate source of therapeutic advice. In preparation for explaining how this learning process will be implemented, the following sections describe related computer-based research and the mechanism by which the system has acquired and used its current knowledge.

## Related Work

Computer-based consultations have been devised for several areas of clinical medicine. Bleich has described a system which offers advice regarding management of electrolyte and acid-base disorders (*2*, *3*), and Sheiner et al. developed a system for assisting in the determination of the correct dosage regimen for achieving desired blood levels of an administered drug (*19*). There are also several systems which have used statistical theory to lead to diagnoses (*7*, *13*, *22*). Sheiner et al. and Bleich chose subject areas in which relevant clinical data tends to be quantitative, and their advice can therefore be verified computationally. When decision making rests heavily on clinical data that are often qualitative and imprecise, however, the advice of a consultation program is evaluated only by acquisition of verifying or contradictory information about the patient. A consultation program is, therefore, acceptable only if it gives correct advice at least as often as does an expert presented with the same diagnostic or therapeutic problems.

The use of Artificial Intelligence (AI) techniques also implies the use and organization of knowledge in a fashion which to a certain extent parallels the reasoning techniques of an intelligent human. Wortman has recently discussed an information processing approach to neurological diagnosis (*25*), and Kulikowski has utilized AI pattern-matching methods in work on a medical diagnostic task (*10*).

More recently Kulikowski has developed a rule-based physiological model of glaucoma which is designed to serve as the basis for diagnostic consultations (*1*, *11*, *12*). The glaucoma consultant draws heavily on AI techniques, representing disease states in terms of linguistic structures and explicitly displaying reasoning procedures. The program described in this report uses related techniques to provide consultations in another area of clinical medicine—infectious disease therapeutics.

## METHODOLOGY

As described above, the focus of this project has been to design a system which is compatible with the physician's own decision making processes. We have not been concerned with developing a model of sequential diagnosis using mathematical and statistical approaches to decision theory. We believe that the best way to encourage a practicing physician to use a consultation system effectively is to convince him that it is reaching decisions in much the same way that he does. The relationship of the current project to AI is embodied in the novel organization of its knowledge, the mechanism by which new decision rules are to be acquired, and the efficient method by which necessary information is accessed and utilized during the consultation itself.

On the basis of frequent discussions (covering approximately two dozen medical records of patients requiring infectious disease therapy), the authors have defined several rules which a rudimentary consultation system needs. These rules have been

SAMPLE RULE

*LISP*

```
PREMISE:
        ($AND   (SAME (VAL CNTXT GRAM) GRAMPOS)
                (SAME (VAL CNTXT MORPH) COCCUS)
                (SAME (VAL CNTXT CONFORM) CHAINS)
                3)
ACTION:
        (CONCLUDE CNTXT IDENT STREPTOCOCCUS TALLY .7)
```

*ENGLISH TRANSLATION*

IF:

THE GRAMSTAIN OF THE ORGANISM IS GRAMPOS, AND THE MORPHOLOGY OF THE ORGANISM IS COCCUS, AND THE GROWTH CONFORMATION OF THE ORGANISM IS CHAINS

THEN:

CONCLUDE THAT THE IDENTITY OF THE ORGANISM IS STREPTOCOCCUS (MODIFIER: THE CERTAINTY TALLY FOR THE PREMISE TIMES .7)

FIG. 1. Coded and translated versions of a sample rule. Note that the ACTION involves a certainty modifier which depends upon both the probability with which the indicated conclusion can be made and the extent to which the items in the PREMISE have been satisfied.

coded, categorized, and hand-entered into the computer program (written in the LISP language (*15*) on a Digital Equipment Corporation (DEC) PDP-10 computer at Stanford Research Institute). Categorization of rules has been by subject matter alone, i.e., some rules permit deductions regarding patient-related data, others discuss the characteristics of infecting organisms, while still others deal with earlier cultures, prior surgical operations, antimicrobial drugs, or therapeutic recommendations. Other than this general categorization, no attempt has been made to relate rules to one another; this is the task of the monitor which drives the consultation program.

An example of one of the rules employed by the system is shown in Fig. 1. All such rules are stored in the machine as compact LISP expressions. However, it was relatively straightforward to write subprograms to translate any rule into an English language version which, although somewhat stilted, is quite adequate for communication with a user unfamiliar with LISP. Both versions of this rule are shown in Fig. 1. All rules consist of a "PREMISE" and an "ACTION." The monitor program examines the PREMISE of a rule and, if the conditions are met within a specified degree of certainty, the consequences of the ACTION portion of the rule may be taken. For example, the sample rule indicates that if gram positive cocci grow in chains then they are apt to be streptococci, i.e., certain characteristics of an organism allow a conclusion to be drawn regarding another of its characteristics—its identity in this case.

In deciding how the rule monitor should function, we looked once again at the way a physician decides upon drug therapy for a patient with an infectious disease. It seems clear that the physician does not have a well structured decision tree stored away which he can call upon to lead him to the best conclusion under all circumstances. Since the physician seeks advice because he wants to know what to prescribe for his patient, the monitor begins by trying to execute the rules that pertain to this goal, i.e., the therapeutic decision rules. As shown in Fig. 2, the monitor program is applied to all of the therapy rules (currently comprising only 15 of the 100) and keeps track of those which come closest to being executed. After testing each of these goal rules (several of which are quickly rejected), the consultation program then chooses the rules in which the PREMISE conditions are best satisfied and recommends drug treatment on the basis of their conclusions. Since the PREMISE of each rule depends upon a certain amount of probabilistic knowledge, the system must look for ways of deducing whether the PREMISE conditions are true for the rule that the monitor is attempting to execute. We have devised a storage system which permits only those rules to be tried which are relevant to the required information—the program does not run through all the rules each time it attempts to draw a conclusion. If no rules are found which permit a specific deduction to be made, then an appropriate question is asked of the physician. Thus, the user is asked to enter information only on subjects about which the system finds it cannot deduce adequate answers. Furthermore, as new rules are added to the program's knowledge base, certain questions which are now being asked of the user will no longer be necessary. This rule-based system design permits new rules to be added without altering any of the old rules—a modular capability which is highly desirable since it facilitates future interactive acquisition of new rules.

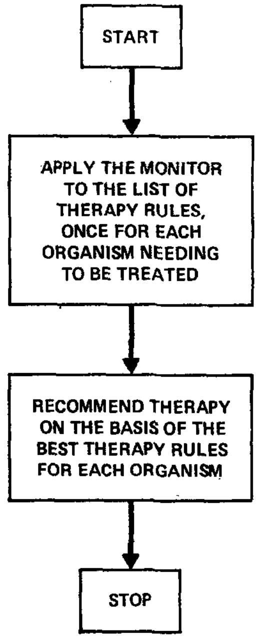


FIG. 2. Flow chart describing the overall organization of the consultation program.

## THE CONSULTATION PROGRAM

A schematic representation of the monitor program which drives the consultation process is shown in Fig. 3. The first rule that the monitor examines is always the initial therapy rule. This rule currently suggests that if the identity of an offending organism is known to be pseudomonas, then gentamicin is the drug of choice. Figure 3 indicates how the monitor deals with this and all other rules. First, the PREMISE is examined (Step 1) to see whether sufficient information is available to answer the question "Is the identity of the organism pseudomonas?" Clearly this question cannot be answered because one piece of necessary information, namely the identity of the offending organism ($=J$), is not known. The monitor therefore retrieves a list ($=Y$) of all rules which may assist in the proper identification of the organism (Step 2), i.e., it retrieves all rules in which conclusions regarding the identity of an organism are mentioned in the ACTION portion of the rule. Step 3 then recursively applies the monitor to each of these potentially useful rules. Eventually the list of rules in $Y$ is exhausted, and the monitor proceeds to Step 4 to decide whether the missing information has now been deduced. Therefore, data are requested directly from the user (Step 5) only if recursive applications of rules have been unable to find the information needed for the PREMISE of the rule. Control then returns to Step 1 and the PREMISE is examined to see if any additional

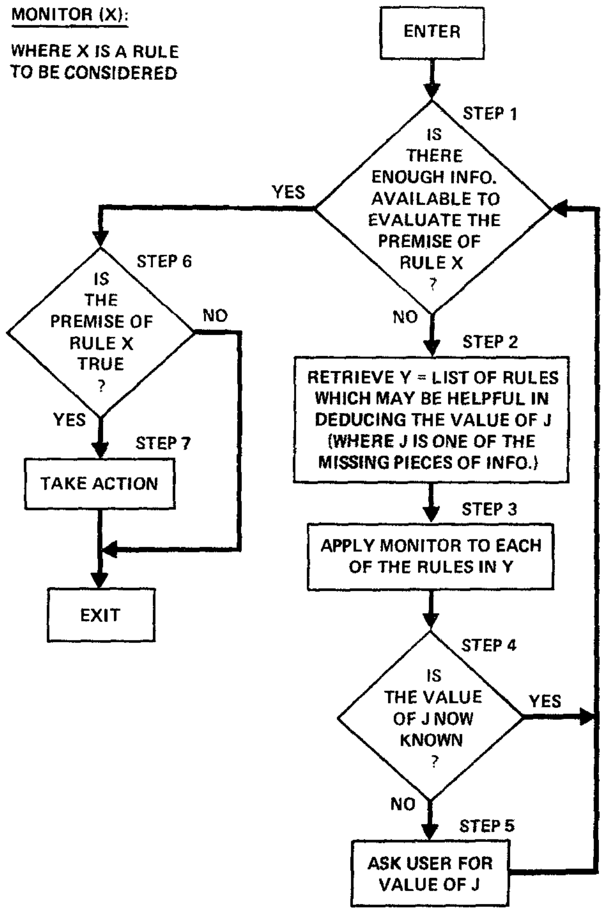


FIG. 3. Flow chart demonstrating the organization and recursive nature of the rule monitor.

information is missing. If not, the truth of the PREMISE is determined (Step 6). If the conditions of the PREMISE have been met, the ACTION portion of the rule is executed (Step 7).

The system described above is capable of using a large number of rules to reach a decision regarding the most appropriate infectious disease therapy for certain clinical situations. An example of one such interactive session between the system and a physician seeking consultative advice is shown in Fig. 4. Every question asked

```
)
INSTRUCTIONS? (Y OR N)
**yes

    I AM HERE TO ADVISE YOU REGARDING AN APPROPRIATE CHOICE OF
    INFECTIOUS DISEASE THERAPY.  I UNDERSTAND THAT YOU HAVE A PATIENT
    FROM WHOM A POSSIBLY POSITIVE CULTURE (CALLED 'CULTURE-1') HAS
    BEEN OBTAINED.  PLEASE ANSWER THE FOLLOWING QUESTIONS,
    TERMINATING EACH RESPONSE WITH 'RETURN'.
    IF YOU ARE NOT CERTAIN OF YOUR ANSWER,  YOU MAY MODIFY THE
    RESPONSE BY INSERTING A CERTAINTY FACTOR (A NUMBER FROM 1 TO 10)
    IN PARENTHESES AFTER YOUR RESPONSE.  ABSOLUTE CERTAINTY (10) IS
    ASSUMED FOR EVERY UNMODIFIED ANSWER.
    NOTE THAT YOU MAY ALSO ENTER '?' IF YOU DO NOT KNOW THE ANSWER
    TO A QUESTION,  '??' IF YOU WOULD LIKE TO SEE A LIST OF
    RECOGNIZED RESPONSES, THE WORD 'WHY' IF YOU WOULD LIKE TO SEE THE
    DECISION RULE WHICH HAS GENERATED THE QUESTION, OR THE WORD
    'HELP' IF YOU ARE CONFUSED BY A QUESTION AND WOULD LIKE IT
    REPHRASED.  TOGETHER WE WILL TRY TO DETERMINE APPROPRIATE THERAPY
    FOR THIS PATIENT.
    SUMMARY:
     '?'    - ANSWER NOT KNOWN
     '??'   - REQUEST FOR RECOGNIZED RESPONSES
     'WHY'  - REQUEST FOR CURRENT DECISION RULE
     'HELP' - REQUEST FOR RESTATEMENT OF QUESTION
    SAMPLE RESPONSE:
       DID ORGANISM-1 GROW IN CLUMPS, CHAINS, OR PAIRS?
      **CHAINS (7) PAIRS (3)


" 2-MAY-73 14:42:55"

                           -----PATIENT-1-----
  PATIENT'S NAME:
  **J. Wilson
  SEX:
  **m
  AGE:
  **53
  LOCATION (WARD):
  **ela
  WHAT IS THE CURRENT INFECTIOUS DISEASE DIAGNOSIS?
  **fuo
  I WILL REFER TO THE MOST RECENT POSITIVE CULTURE FOR WHICH YOU
  DESIRE THERAPEUTIC ADVICE AS:

                           -----CULTURE-1-----
     FROM WHAT SITE WAS THE SPECIMEN FOR CULTURE-1 TAKEN?
     **blood
     HOW MANY DAYS AGO WAS THIS BLOOD CULTURE (CULTURE-1) OBTAINED?
     **2
     I WILL REFER TO THE FIRST OFFENDING ORGANISM FROM THIS BLOOD
     CULTURE (CULTURE-1) AS:

                           -----ORGANISM-1-----
        ENTER THE IDENTITY (GENUS) OF ORGANISM-1:
        **?
        IS ORGANISM-1 A ROD OR COCCUS (ETC.):
        **rod
```

FIG. 4. A sample consultation based upon the initial 100 rules. Entries by the user are underlined and always follow a double asterisk.

```
          STAINING CHARACTERISTICS OF ORGANISM-1 (GRAM):
          **neg
          DID ORGANISM-1 GROW IN THE AEROBIC BOTTLE?
          **y
          DID ORGANISM-1 GROW IN THE ANAEROBIC BOTTLE?
          **n
          DOES THE PATIENT HAVE 'PSEUDOMONAS-TYPE' HEMORRHAGIC SKIN
          LESIONS?
          **n
          INCLUDING THIS CULTURE, HOW MANY BLOOD CULTURES WERE DRAWN
          ON THE SAME DAY?
          **2
          HOW MANY OF THE BLOOD CULTURES WERE POSITIVE FOR ORGANISM-1?
          **2
          WHAT IS THE SUSPECTED NON-STERILE PORTAL OF ENTRY OF
          ORGANISM-1 INTO THE BLOOD?
          **gi
PLEASE WAIT WHILE I CHECK MY FILES FOR A MOMENT...
          DO YOU CONSIDER J. WILSON (PATIENT-1) TO BE A COMPROMISED
          HOST? (ALCOHOLIC, IMMUNOSUPPRESSED, ETC.):
          **y
          WAS THE INFECTION WITH ORGANISM-1 ACQUIRED SINCE THE PATIENT
          WAS ADMITTED?
          **n
          HAS J. WILSON (PATIENT-1) HAD ANY OPERATIONS WHICH YOU FEEL
          MAY BE RELEVANT TO THE CURRENT INFECTION?
          **n
          HAVE THERE BEEN ANY OTHER POSITIVE CULTURES FROM THIS OR ANY
          OTHER SITE WITHIN THE LAST APPROXIMATELY 30 DAYS?
          **y

                         -----CULTURE-2-----
            FROM WHAT SITE WAS THE SPECIMEN FOR CULTURE-2 TAKEN?
            **throat
            HOW MANY DAYS AGO WAS THIS THROAT CULTURE (CULTURE-2)
            OBTAINED?
            **5
            DO YOU HAVE IDENTIFYING INFORMATION ON THE ORGANISM THAT
            WAS ISOLATED FROM THIS THROAT CULTURE (CULTURE-2)?
            **y

                         -----ORGANISM-2-----
              ENTER THE IDENTITY (GENUS) OF ORGANISM-2:
              **serratia
            ANY OTHER SIGNIFICANT ORGANISMS ISOLATED FROM THIS THROAT
            CULTURE (CULTURE-2)?
            **n
          ANY OTHER SIGNIFICANT PRIOR CULTURES FROM WHICH PATHOGENS
          WERE ISOLATED?
          **n
          WAS THE INFECTION WITH THE SERRATIA (ORGANISM-2) ACQUIRED
          SINCE THE PATIENT WAS ADMITTED?
          **n
          WAS THE PATIENT TREATED FOR THE INFECTION WITH THE SERRATIA
          (ORGANISM-2)?
          **n
```

Fig. 4—*Continued*

```
        WAS ANY ADDITIONAL CHEMOTHERAPY RECENTLY BEGUN TO COMBAT
        ORGANISM-1?
        **n
     ANY OTHER ORGANISMS ISOLATED FROM THIS BLOOD CULTURE
     (CULTURE-1) FOR WHICH YOU WOULD LIKE A THERAPEUTIC
     RECOMMENDATION?
     **n
  WERE ANY OTHER POSITIVE SPECIMINS RECENTLY DRAWN FROM J. WILSON
  (PATIENT-1) WHICH REVEALED ADDITIONAL ORGANISMS FOR WHICH YOU WISH
  THERAPEUTIC ADVICE?
  **n


    CONCLUSIONS REGARDING ORGANISMS FOR WHICH THE PATIENT IS TO
         BE TREATED...
(ORGANISM-1)
    THE IDENTITY OF ORGANISM-1:
         MOST LIKELY:  SERRATIA OR PSEUDOMONAS
         CONSIDERED BUT RULED OUT:  PROTEUS OR KLEBSIELLA OR E.COLI OR
                                    STREPTOCOCCUS OR CLOSTRIDIUM OR
                                    BACTEROIDES
    THE CLASS OF ORGANISM-1:
         MOST LIKELY:  ENTEROBACTERIACEAE
    AN ORGANISM WHICH MAY BE THE SAME AS ORGANISM-1:
         MOST LIKELY:  ORGANISM-2


TURN TO NEW PAGE - HIT 'RETURN'
```

Fig. 4—*Continued*

```
MY THERAPY RECOMMENDATION WILL BE BASED ON THE FOLLOWING CONDITIONS
TAKEN FROM THE RELEVANT THERAPY RULES:
      <1>      THE IDENTITY OF ORGANISM-1 IS SERRATIA
               (RULE077 - 7/10)
      <2>      THE IDENTITY OF ORGANISM-1 IS NOT KNOWN WITH CERTAINTY,
             AND
             THE IDENTITY OF ORGANISM-1 IS PSEUDOMONAS
               (RULE104 - 7/10)

   IS THE SERRATIA (ORGANISM-2) SENSITIVE TO GENTAMICIN?
   **yes


[REC 1] MY PREFERRED THERAPY RECOMMENDATION IS AS FOLLOWS:
     IN ORDER TO COVER FOR <1> <2>:
        GIVE: GENTAMICIN  (8/10)
             DOSE: 1.5 MG/KG Q8H - IV OR IM
             COMMENTS: MODIFY DOSE IN RENAL FAILURE

DO YOU WISH TO SEE THE NEXT CHOICE?
**yes


   IS THE SERRATIA (ORGANISM-2) SENSITIVE TO KANAMYCIN?
   **?

[REC 2] NEXT BEST THERAPY RECOMMENDATION:
     IN ORDER TO COVER FOR <1>:
        GIVE: KANAMYCIN  (6/10)
             DOSE: 15 MG/KG Q12H - IM
             COMMENTS: MODIFY DOSE IN RENAL FAILURE
     IN ORDER TO COVER FOR <2>:
        GIVE: COLISTIN  (5/10)
             DOSE: 1.25 MG/KG Q6H - IM
             COMMENTS: MODIFY DOSE IN RENAL FAILURE

DO YOU WISH TO SEE THE NEXT CHOICE?
**yes


[REC 3] NEXT BEST THERAPY RECOMMENDATION:
     IN ORDER TO COVER FOR <2>:
        GIVE: POLYMYXIN  (5/10)
             DOSE: 1.0 MG/KG Q8H - IM

DO YOU WISH ADVICE ON ANOTHER PATIENT?
**no

THANK-YOU
_
```

Fig. 4—*Continued*

by the program corresponds to a point at which Step 5 in Fig. 3 is reached because the requested information could not be deduced from previously acquired data.

Note that the user also has four special options available to him whenever a question is asked. Any question may be answered with a single question mark ("?") if the requested information is not known. The program then simply proceeds and gives the best advice it can despite the missing information. Entry of the word "HELP" tells the program to rephrase the question and also to display a list of recognized responses. A double question mark ("??") requests only the recognized response list.

A fourth option allows the system to serve a tutorial as well as consultative role. As shown in Fig. 3, whenever a question is asked of the user, the rule which the program is attempting to evaluate (and which has sent the monitor from Step 1 to Step 5) is known to the program. Thus, if the physician is puzzled by a question and wants to know why the machine is asking for the requested piece of information, he enters the word "WHY" and the program prints out an English language translation of the rule that it is attempting to execute. Figure 5 shows a portion of a consultation that is similar to the one shown in Fig. 4 but in which the "WHY" option and the others described above have been utilized.

The programs that drive the consultation process are expected to require minimal refinements in the future. Although the mechanisms for relating existing rules to one another and for using them appropriately are already adequate, the system should be capable of providing advice of a higher quality and broader scope as the number of rules is increased and existing rules are modified. The interactive process is very fast because the user's answers generally require minimal analysis. Furthermore, rules are coded as LISP expressions which may be evaluated directly without time-consuming syntactic parsing. An average consultation using the current system requires approximately five minutes of terminal time and involves a maximum pause of three seconds between the user's response to one question and the initiation of the following prompt.

## Knowledge Acquisition Program

A model for acquiring new rules has been well defined, but only the first stage in the implementation of this capability has been completed. Experts in the treatment of infectious diseases will be asked to come to the computer terminal with the medical record of a patient whose disease has presented a difficult antimicrobial therapeutic problem. The expert will then ask the system for advice via the consultation program, challenging its limited knowledge in much the same way that a novice student is first challenged by an expert when he begins to see patients on the wards. The expert will answer questions about the patient when they are asked by the program and will take note of irrelevant, inappropriate, or missing questions. At the termination of the consultation session, the expert will critically evaluate the

```
                              .
                              .
                              .
                              .
                              .
                              .
I WILL REFER TO THE FIRST OFFENDING ORGANISM FROM THIS BLOOD
CULTURE (CULTURE-1) AS:

                    -----ORGANISM-1-----
   ENTER THE IDENTITY (GENUS) OF ORGANISM-1:
   **??
   EXPECTED RESPONSES ARE: CANDIDA ACTINOMYCETES ARIZONA
   BACTEROIDES ETC.
   ENTER THE IDENTITY (GENUS) OF ORGANISM-1:
   **?
   IS ORGANISM-1 A ROD OR COCCUS (ETC.):
   **rod
   STAINING CHARACTERISTICS OF ORGANISM-1 (GRAM):
   **neg
   DID ORGANISM-1 GROW IN THE AEROBIC BOTTLE?
   **y
   DID ORGANISM-1 GROW IN THE ANAEROBIC BOTTLE?
   **WHY
IF:  THE SITE OF THIS BLOOD CULTURE IS BLOOD, AND
     THE IDENTITY OF ORGANISM-1 IS NOT KNOWN WITH CERTAINTY,
     AND
     AEROBIC BOTTLE GROWTH BY ORGANISM-1 HAS BEEN INFERRED OR
     DEMONSTRATED, AND
     ANAEROBIC BOTTLE GROWTH BY ORGANISM-1 HAS BEEN INFERRED
   ) OR DEMONSTRATED
THEN:  DO THE FOLLOWING:,
       CONCLUDE THAT THE AEROBICITY OF ORGANISM-1 IS FACUL
       (MODIFIER: THE CERTAINTY TALLY FOR THE PREMISE TIMES .9) ,

       CONCLUDE THAT THE AEROBICITY OF ORGANISM-1 IS
       ANAEROBIC (MODIFIER: THE CERTAINTY TALLY FOR THE PREMISE
       TIMES .2)
   DID ORGANISM-1 GROW IN THE ANAEROBIC BOTTLE?
   **n
   DOES THE PATIENT HAVE 'PSEUDOMONAS-TYPE' HEMORRHAGIC SKIN
   LESIONS?
   **n
   INCLUDING THIS CULTURE, HOW MANY BLOOD CULTURES WERE DRAWN
   ON THE SAME DAY?
   **1
   WHAT IS THE SUSPECTED NON-STERILE PORTAL OF ENTRY OF
   ORGANISM-1 INTO THE BLOOD?
   **HELP
   I NEED TO KNOW THE PORTAL OF ENTRY OF ORGANISM-1
   EXPECTED RESPONSES ARE: THROAT VAGINA RESP SKIN GI
   WHAT IS THE SUSPECTED NON-STERILE PORTAL OF ENTRY OF
   ORGANISM-1 INTO THE BLOOD?
   **gi
                              .
                              .
                              .
                              .
                              .
```

FIG. 5. A portion of a dialog which demonstrates the usefulness of the special options available to the user of the consultation program ('?', '??', 'HELP', and 'WHY').

interactive process to identify weaknesses in the logic and content. A special teaching mode will be available which will permit the user/teacher to query the program regarding its knowledge of specified medical topics, why it asked certain questions in the consultation process, and which inference rules were used to reach specific decisions. Once the expert has identified the source of errors in the consultation, he will either modify old rules or enter new ones. The system will accept new or modified rules, translate them into the proper internal form, check them for inconsistencies with currently existing rules, and store noncontradictory new rules for use during future runs of the consultation program.

Essential to the success of the system is the requirement that experts in infectious disease therapy should not be required to learn a computer language in order to enter new rules. Although some training will no doubt be necessary, and although the expert may need some knowledge of the overall organization of the consultation program, we are eager to permit most of the learning dialog to proceed in natural language. Our limitations here are both the current state of the art of computational linguistics (*18*, *23*, *24*) plus our goal of developing an operational prototype system as soon as possible. Our objective, then, is to develop a system which understands a subset of natural language within the context of our limited world model and which can accomplish the following tasks:

(1) Answer questions about the data base of rules—i.e., be able to discuss current knowledge of the system;

(2) Answer questions about a consultation concerning a specific patient—i.e., maintain a functional memory of the preceding conversation.

(3) Accept natural language sentences for translation into LISP expressions which convey their meaning to the monitor—i.e., "learn" new rules for use in future consultations.

We have begun this part of the project by implementing a module capable of undertaking task 1 from the list above. Several question-answering (QA) approaches have been developed in recent years (*16*, *20*, *21*). We have been able to make useful simplifying assumptions in our QA program because of the straightforward structures to which our rules conform. Green's BASEBALL program (*8*) demonstrated the validity of such an approach when the data structures and question formats are predictable. Our QA program accepts natural language questions about the system's knowledge of infectious disease chemotherapy, translates the question into a specifier which clarifies the format of any rule(s) which could relate to the question, and retrieves all rules satisfying the specifier format. Since all system knowledge is contained in its decision rules, question-answering can be effectively handled by simple rule retrieval. At present we merely print out the English translation of the relevant rules (see Fig. 6) and expect the user to derive the answer to his question from the printed rule. An extension of the module will be to extract the specific piece of information itself and to answer accordingly with a simple sentence.

### The Question-Answering Module

** Is either chloramphenicol or kanamycin okay for treatment of salmonella infections?

YES.

*RULE083*

IF:

THE IDENTITY OF THE ORGANISM IS SALMONELLA

THEN:

I RECOMMEND (IN ORDER OF PREFERENCE):
1-CHLORAMPHENICOL
2-AMPICILLIN
3-CEPHALOTHIN

** How do you know if an organism is a bacteroides?

*RULE030*

IF:

THE GRAMSTAIN OF THE ORGANISM IS GRAMNEG, AND THE MORPHOLOGY OF THE ORGANISM IS ROD, AND THE AEROBICITY OF THE ORGANISM IS ANAEROBIC

THEN:

CONCLUDE THAT THE IDENTITY OF THE ORGANISM IS BACTEROIDES (MODIFIER: THE CERTAINTY TALLY FOR THE PREMISE TIMES .6)

FIG. 6. Two examples of the kinds of questions which the QA module is able to handle.

We anticipate that the 700 word dictionary compiled for the QA module will not need major extension in order to handle most of the situations arising during the implementation of tasks 2 and 3. The dictionary is now simply a synonym lookup table combined with a mechanism for deducing what characteristics of organisms, patients, cultures, etc., the question is referring to. At present the system does no syntactic processing of input questions, and needs only minimal semantic information. Because most terms in this domain have only one meaning, we need not concern ourselves with complicated approaches to disambiguation.

## Summary

The antimicrobial therapy consultation system utilizes a flexible representation of knowledge which facilitates interactive advice-giving sessions with physicians. Ideas for acquisition, storage, and utilization of the knowledge of the task domain, plus considerations of natural language processing, draw upon current Artificial Intelligence research. The feasibility of the judgmental rule approach used here has been demonstrated with a limited knowledge base of approximately 100 rules. The system is self-explanatory, easy-to-use, and designed for instruction as well as consultation. Its ultimate success as a clinically useful tool depends upon acquisition

of additional rules and thus upon cooperation of infectious disease experts willing to improve the program's knowledge base.

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